

Project Green-PVC	Client Mundra Petrochem Ltd.	Plant Code MPL-OEPMC	Document Identification Number OEPMC-EQS-0000-EC-00008	Contract No. 66-6753
	TECHNICAL SPECIFICATION FOR THERMAL INSULATION			
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1 SCOPE

- 1.1 This specification covers the requirements for above ground, external insulation of piping and equipment. It contains generally applicable requirements regarding insulation materials and its application.
- 1.2 In case of conflict between this specification and other related specifications, vendor shall notify Owner / OE / PMC.
- 1.3 Any deviation from this specification shall be agreed upon in writing.

2 GENERAL

- 2.1 Uhde's "Erection - Thermal Insulation" (UN 9256-01 Part 1(M)) and "Insulation – Heat and cold Insulation – Arrangement details" (UN 5001-02 Part 1(M)) forms an integral part of this specification along with the specific requirements for the Project. If discrepancies are found among this Technical Specification for thermal insulation, UN 9256-01 Part 1(M) and UN 5001-02 Part 1(M), then following shall apply -

Any information given in Technical Specification for thermal insulation excluding UN 9256-01 Part 1(M) and UN 5001-02 Part 1(M) will supersede any information in UN 9256-01 Part 1(M) and UN 5001-02 Part 1(M).

Any information given in Clause 3.0 viz. "Specific Requirements for the project" will supersede any information given in any other part of this Technical Specification.

- 2.2 Insulation material shall be delivered to site with proper packing such as polythene lined fabric etc. Insulation showing any evidence of moisture shall be rejected. All delivered material shall be kept dry and stored in dry weatherproof shelter. The storage of material at site is completely contractor's responsibility.
- 2.3 Insulation Supports on Equipment
Insulation supports in the form of rings, lugs etc. requiring welding on vessels, tanks etc. will be provided by the Owner/Equipment Vendor. All other insulation supports, spacer rings etc. shall be supplied, fabricated and installed by the insulation contractor as a part of insulation work. Insulation supports for vertical equipment shall be as per list given below (not in scope of insulation contractor).

	1. Hot Insulation / Hot + Fire Protection Insulation/	: UN 2004-06 (M), Page 1.
	Personal protection + Fire Protection/ Only Fire Protection	
	2. Cold Insulation / Anti-Sweat Insulation	: UN 2004-06 (M), Page 2.
	3. Personal Protection (Option 2)	: UN 2004-06 Part 1 / 2 / 3.
	Personal Protection (Option 1)	: UN 2004-06 (M), Page 1.
	4. Cold + Fire Protection Insulation	: UN 2004-06 (M), Page 2.

Horizontal vessels shall not be provided any insulation supports by the Owner, refer Clause 3.3.3 for Insulation of Horizontal Vessels.

Once the equipment is handed over for application of insulation work, no welding is allowed for any reason.

- 2.4 In case of equipment having cold insulation, wherever there are protrusions in the form of support skirts, leg supports etc. the insulation shall be continued for a distance equal to five times the insulation thickness.
- 2.5 Scaffolding
It will be insulation contractor's responsibility to provide all necessary scaffoldings, required for all equipment, piping etc. The rates shall be inclusive of providing scaffolding and safety equipment at all heights.

Noted

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3 SPECIFIC REQUIREMENTS OF THE PROJECT

3.1 Painting of surfaces to be insulated

Carbon steel and low alloy steel piping and equipment shall be protected against corrosion by painting as per Owner's / OE's / PMC's Painting Specification. Unless otherwise specified, painting is not in the scope of insulation contractor. All other surfaces of piping and equipment namely stainless steel, Monel etc. shall not be painted before application of Insulation.

3.2 Inspection

3.2.1 All insulation materials shall be subjected to inspection by Owner / OE / PMC before application. In case of doubt, Owner's / OE's / PMC's representative will have liberty to get the material tested by the contractor at any approved test laboratory. Any material not meeting the specified requirement will be rejected.

3.2.2 In case of Mineral Wool, all the tests as specified in relevant IS specification shall be performed and the special requirements as specified in the Technical specification should be fulfilled. Some of the properties which should be tested are given below:

1. Density
2. Thermal Conductivity
3. Chloride content
4. Sulphur content
5. Moisture Absorption
6. Shot content
7. Recovery after compression
8. Heat Resistance

3.2.3 In case of foams also, all the tests shall be performed as per the standards specified in the Technical Specification. Some of the important properties which should be tested are as given below:

1. Density
2. Thermal conductivity
3. Water vapour permeability and water absorption
4. Flammability test
5. Compressive strength
6. Closed cell content

3.2.4 Every batch of materials supplied should be backed by test certificates. Results of inspection and tests will be recorded by the contractor in the Inspection Reports Pro forma which will be approved by Site-in-Charge / Owner / OE / PMC. These reports as well as test certificates will be submitted as part of completion certificate.

3.2.5 Installation of insulation is also subject to inspection at all times by Site-in-Charge. Insulation found to be improperly installed shall be removed and re-installed properly. The contractor shall carry out all instructions given during inspection and shall ensure that the work is being carried out according to the Technical Specification.

3.2.6 Tests for thickness shall be carried out after application. Thickness at any point shall not be less than the indicated thickness and shall not be more than 110% of the specified thickness.

3.2.7 Inspection of materials and/or installation by Owner / OE / PMC shall not relieve the contractor of his responsibility to ensure that finished insulation conform to specified requirements and is free from defects.

3.2.8 The contractor shall provide for the purpose of inspection, access ladders, lighting and necessary instruments at his own cost.

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3.3 Application

3.3.1 Thickness of insulation shall be as specified in the Insulation schedule / Isometric drawings prepared for equipment and piping or process and instrument diagram. Whenever the thickness is not indicated, the same shall be taken from Insulation thickness table (Refer Clause 3.9 below) with notification to Owner / OE / PMC. In case of any conflict, contractor shall notify Owner / OE / PMC and ask for necessary clarification before starting application work.

3.3.2 Extent of Insulation :

3.3.2.1 Unless specifically indicated, the insulation work shall be performed as per Isometric drawings prepared for piping and equipment.

3.3.2.2 In case of cold insulation, the Body flanges of all the equipment and valves and flange joints in piping shall be insulated. This is required to avoid formation of ice, white frost or condensation during operation.

3.3.2.3 In case of personnel protection, all the flanges and valves shall be insulated.

3.3.2.4 In case of hot insulation also, the body flanges of equipment as well as valves and flanges in piping shall be insulated for heat conservation unless specifically indicated.

3.3.2.5 Whenever external stiffening rings are provided on the equipment, the same shall be insulated with the thickness same as that of equipment.

3.3.2.6 Nameplates, portion above packing glands on valves, portion above the housing flanges of safety valves and swivel and expansion joints/expansion bellow shall not be insulated.

3.3.3 Insulation of Horizontal Equipment :

For insulation of Shell of Horizontal equipment :

Horizontal equipment shell shall be insulated with the use of spacers as per Figure.1 of insulation standard, UN 5001-02 Part 1(M) which is indicated as : Heat Insulation – Spacer for Horizontal Pipes. This avoid any welded cleats on the horizontal Vessel and the insulation contractor uses the loose clips as indicated in the standard.

For insulation of dished end of Horizontal equipment following procedure shall be followed:

1. Making of small strip ring for dish end and big ring at junction of shell & dish part and same are joined /connected with strips (Tie strips from Big to small ring).
2. LRB (Light resin bonded) matters is fixed over this strip ring.
3. Cladding sheets are fixed with screws in shapes.

3.4 Measurement of Work

Basis of measurement for all types of insulation work shall be same.

It shall be done on a unit rate basis for the piping items, i.e. pipe length in meter and fittings, flanges, valves in numbers.

The measurement of insulation area of equipment shall be performed as per IS 14164.

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3.5 Schedule of Quantity

- 3.5.1 The information to be supplied to contractor for obtaining the rates of insulation shall be given in document called "Schedule of Quantity".
- 3.5.2 In the schedule of quantity for equipment, the list of equipment to be insulated is given with details as mentioned below:
- Type of Insulation
 - Dimensions of Equipment
 - Type of Equipment - Horizontal or Vertical
 - Material of Construction
 - Operating Temperature
 - Thickness of Insulation
 - Area of Insulation – for reference
- 3.5.3 In case of Schedule of Quantity for Piping, the same is given separately.
- Carbon and Low alloy steel piping.
 - Stainless steel / non-ferrous piping.

In the schedule of quantity, respective size pipe length having same insulation thickness is indicated. Quantities of piping elbows, tees, bends, valves, flanges etc. are listed separately with respect to insulation thickness.

3.6 Guarantee

- 3.6.1 The guarantee for Thermal conductivity values of various insulation materials as indicated in the specification is in contractor's scope.
- 3.6.2 The guarantee for insulation material and workmanship is also included in insulation contractor's scope.



3.7 Scope of supply

The scope of work will consist of but not limited to following:

Supply of all types of insulation as well as ancillary materials and application of insulation of all types mentioned below on equipment like Columns, Vessels, Heat Exchangers, Tanks etc. and piping, fittings, valves, flanges etc.

1. Hot Insulation
2. Insulation for Personnel Protection (Option 1 : Resin Bonded Rockwool / Option 2 : Metal Guards as per UN 2004-06 Part 1,2 & 3)
3. Cold Insulation / Anti-Sweat Insulation (Note-1)
4. Cold + Hot Insulation
5. Only Fire Protection Insulation (Note-5)
6. Hot + Fire Protection insulation (Note-2)
7. Cold + Fire Protection insulation (Note-3)
8. Personal Protection + Fire Protection insulation (Note-4)

Notes :

- 1) Cold insulation is to be applied for the cases wherever Anti-sweat insulation is specified. Accordingly, Insulation Thicknesses shall be followed same as per 'Cold Insulation Table' from Clause 3.9. Also, specifications/ standards applicable for Cold Insulation shall be followed for Anti-sweat Insulation cases.
- 2) For Hot + Fire protection insulation application, thickness shall be followed same as per 'Hot Insulation Table' from Clause 3.9. Also, specifications/ standards applicable for Hot Insulation shall be followed for the same.

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- 3) For Cold + Fire Protection insulation application, insulation thickness shall be followed same as per respective Table from Clause 3.9. Also, specifications/ standards applicable for the same shall be followed from Appendix-A5.
- 4) For Personal Protection + Fire Protection insulation application, insulation thickness shall be followed same as per 'Personal Protection Insulation Table' from Clause 3.9. Also, specifications/ standards applicable for Hot Insulation shall be followed for Personal Protection + Fire Protection Insulation cases, along with the special notes for Fire cases.
- 5) For Only Fire Protection insulation application, **insulation thickness = 40mm** to be considered. Also, specifications/ standards applicable for Hot Insulation shall be followed for Only Fire Protection Insulation cases, along with the special notes for Fire cases.



3.8 Insulating materials and other hardware requirement

- 1) Various acceptable materials for different types of insulation are detailed in the Appendices enclosed. Vendor must confirm compliance with them.
- 2) Materials shall generally be in accordance with ASTM or equivalent standards and shall pass the inspection specified in the applicable standard before use.
- 3) Preferred make for Foam Glass (Cellular Glass) is Pittsburgh Corning with their own adhesive material.
- 4) For cavity filling & protecting foam glass, site uses POP & Gypsum powder mixture, on top of Foam Glass.
- 5) Undertaking shall be taken from vendor on Non-Asbestos use.
- 6) Insulation Packing material shall be free from Thermocol. Environment friendly material shall be used.
- 7) In case of Fire protection insulation applications, incombustible materials shall be used as insulating materials. Insulating materials used to insulate piping / equipment, containing combustible liquids that may auto-oxidize and self-heat shall be of high-quality fireproofing properties (as per API 521 requirements) in addition to good insulation properties. This requirement shall be necessarily and sufficiently complied and ensured.
- 8) Fire insulation, or insulation that is part of a composite system, should be capable of withstanding an exposure temperature of approximately 904 °C (1660 °F) for up to 2 h. Insulation system materials selection should consider equipment metallurgy while providing required jacket integrity at fire water pressures and fire temperatures.

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3.9 Insulation Tables for Reference as follows :

INSULATION TABLE FOR COLD INSULATION / ANTI-SWEAT INSULATION (PUF)

Nominal size		Insulation Thickness For Temperatures °C											
Inch	mm	20 °C	10 °C	0 °C	- 10 °C	- 20 °C	- 30 °C	- 40 °C	- 50 °C	- 60 °C	- 70 °C	- 80 °C	- 90 °C
1/2	15	30	30	30	30	30	30	40	40	40	50	50	50
3/4	20	30	30	30	30	30	40	40	40	40	50	50	50
1	25	30	30	30	30	30	40	40	40	50	50	50	50
1 1/4	32	30	30	30	30	30	40	40	40	50	50	50	60
1 1/2	40	30	30	30	30	30	40	40	50	50	50	50	60
2	50	30	30	30	30	40	40	40	50	50	50	60	60
2 1/2	65	30	30	30	30	40	40	40	50	50	60	60	60
3	80	30	30	30	30	40	40	50	50	50	60	60	60
3 1/2	95	30	30	30	30	40	40	50	50	60	60	60	70
4	100	30	30	30	30	40	40	50	50	60	60	70	70
5	125	30	30	30	40	40	40	50	50	60	60	70	70
6	150	30	30	30	40	40	50	50	60	60	70	70	70
8	200	30	30	30	40	40	50	50	60	60	70	70	70
10	250	30	30	30	40	50	50	60	60	60	70	70	80
12	300	30	30	30	40	50	50	60	60	70	70	80	80
14	350	30	30	30	40	50	50	60	60	70	70	80	80
16	400	30	30	40	40	50	60	60	60	70	70	80	80
18	450	30	30	40	40	50	60	60	70	70	80	80	80
20	500	30	30	40	40	50	60	60	70	70	80	80	80
24	600	30	30	40	40	50	60	60	70	70	80	80	100
30	750	30	30	40	50	50	60	60	70	70	80	100	100
36	900	30	30	40	50	50	60	60	70	70	80	100	100
48	1200	30	30	40	50	50	60	60	70	70	80	100	100
> 48	> 1200	30	30	40	50	50	60	60	70	70	80	100	100

Note: All thickness in mm only.

Basis: Ambient temperature : 47.8°C (Max.)
Relative humidity : 80% at DBT 40°C
Wind velocity : 8 kmph

Noted

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Insulation material : Polyurethane foam as per IS 12436 (Density - 34 kg/m³), pre-formed pipe sections.

Cladding material : Aluminium

Mean thermal conductivity values considered from ASTM curves

No. of layers of insulation : First layer upto 50 mm
Second layer >50 mm to 100 mm
Third layer >100 mm

Heat loss Max. 40 kcal/m²hr

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INSULATION TABLE FOR COLD + FIRE PROTECTION INSULATION (Cellular Glass)

Nominal size		Insulation Thickness For Temperatures °C											
Inch	mm	20 °C	10 °C	0 °C	- 10 °C	- 20 °C	- 30 °C	- 40 °C	- 50 °C	- 60 °C	- 70 °C	- 80 °C	- 90 °C
1/2	15	40	40	40	40	40	40	60	60	60	70	70	70
3/4	20	40	40	40	40	40	60	60	60	60	70	70	70
1	25	40	40	40	40	40	60	60	60	70	70	70	70
1 1/4	32	40	40	40	40	40	60	60	60	70	70	70	80
1 1/2	40	40	40	40	40	40	60	60	70	70	70	70	80
2	50	40	40	40	40	60	60	60	70	70	70	80	80
2 1/2	65	40	40	40	40	60	60	60	70	70	80	80	80
3	80	40	40	40	40	60	60	70	70	70	80	80	80
3 1/2	95	40	40	40	40	60	60	70	70	80	80	80	100
4	100	40	40	40	40	60	60	70	70	80	80	100	100
5	125	40	40	40	60	60	60	70	70	80	80	100	100
6	150	40	40	40	60	60	70	70	80	80	100	100	100
8	200	40	40	40	60	60	70	70	80	80	100	100	100
10	250	40	40	40	60	70	70	80	80	80	100	100	110
12	300	40	40	40	60	70	70	80	80	100	100	110	110
14	350	40	40	40	60	70	70	80	80	100	100	110	110
16	400	40	40	60	60	70	80	80	80	100	100	110	110
18	450	40	40	60	60	70	80	80	100	100	110	110	110
20	500	40	40	60	60	70	80	80	100	100	110	110	110
24	600	40	40	60	60	70	80	80	100	100	110	110	140
30	750	40	40	60	70	70	80	80	100	100	110	140	140
36	900	40	40	60	70	70	90	90	100	100	110	140	140
48	1200	40	40	60	70	70	90	90	100	100	110	140	140
> 48	> 1200	40	40	60	70	70	90	90	100	100	110	140	140

Note: All thickness in mm only.

Basis: Ambient temperature : 47.8°C (Max.)
Relative humidity : 80% at DBT 40°C
Wind velocity : 8 kmph

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Insulation material : Cellular Glass (Foam Glass) as per ASTM 552, E136 & E228 (Density – 98 - 139 kg/m³), pre-formed pipe sections.

Cladding material : Galvanized Steel (For Fire Protection)

Mean thermal conductivity values considered from ASTM curves.

Flammability : Non-combustible (ASTM E 136 or DIN 4102 or BS 476)

Heat loss : Max. 40 kcal/m²hr

Noted

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STANDARD INSULATION TABLE FOR HOT INSULATION /



HOT + FIRE PROTECTION INSULATION (Resin Bonded Rockwool)

Nominal size		Insulation Thickness For Temperatures ° C upto							
Inch	mm	100 °C	150 °C	200 °C	250 °C	300 °C	350 °C	400 °C	450 °C
1/2	15	30	40	50	60	70	90	100	120
3/4	20	30	40	50	60	80	90	110	120
1	25	30	40	50	60	80	100	110	130
1 1/4	32	30	40	50	70	80	100	120	140
1 1/2	40	30	40	60	70	90	100	120	140
2	50	30	40	60	70	90	110	130	150
2 1/2	65	30	40	60	80	90	110	130	150
3	80	30	50	60	80	100	120	140	160
3 1/2	95	30	50	60	80	100	120	140	160
4	100	30	50	70	80	100	120	150	170
5	125	30	50	70	90	110	130	150	180
6	150	30	50	70	90	110	130	160	180
8	200	30	50	70	90	120	140	170	190
10	250	30	50	70	100	120	150	170	200
12	300	30	50	80	100	120	150	180	210
14	350	40	60	80	100	130	150	180	210
16	400	40	60	80	100	130	160	190	220
18	450	40	60	80	100	130	160	190	220
20	500	40	60	80	100	130	160	190	230
24	600	40	60	80	110	140	170	200	230
30	750	40	60	80	110	140	170	210	240
36	900	40	60	80	110	140	170	210	250
48	1200	40	60	80	110	140	180	220	260
Above 48	> 1200	40	60	80	110	140	180	220	260

Note: All thickness in mm only.

Basis: Ambient temperature : 5°C
Wind velocity : 8 kmph
Insulation material : Resin bonded rockwool as per IS-8183 for mattresses and IS-9842 for preformed pipe sections.
Cladding material : Aluminium (For Hot Insulation)
Galvanized Steel (For Hot + Fire Protection Insulation)

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Mean thermal conductivity values considered from IS-8183 (for material density 120 - 150 kg/m³)

05	Conductivity (mw/cm°C at external fire temperature of 904°C) : not higher than 5.77.
	Flammability (Fire Protection case) : Non-combustible (ASTM E 136 or DIN 4102 or BS 476)
	The insulating materials and adhesives, which will be used, shall be certified as non-combustible (Class 0-25 to ASTM E84 or equivalent).

No. of layers of insulation : First layer upto 75 mm
Second layer >75 mm to 150 mm
Third layer >150 mm

Heat loss : Max. 100 kcal/m²/hr (Calculated using following equation)

$$(T_p - T_s) / [r_2 \times (\ln r_2 / r_1) / K_m] = 0.548 \times \epsilon \times [(T_s / 55.55)^4 - (T_a / 55.55)^4] \times 1.957 (t_s - t_a)^{1.25} \times (1 + 2.857 V)^{0.5}$$

T_p : Process Temperature, °K T_s : Surface Temperature, °K
T_a : Ambient Temperature, °K t_s : Surface Temperature, °C
t_a : Ambient Temperature, °C r₁ : Outer radius of pipe, in mts
r₂ : Outer radius of pipe including insulation, in metres
k_m : Mean thermal conductivity W/m°C
ε : Surface emittance
V : Wind velocity, m/s

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**STANDARD INSULATION TABLE FOR PERSONNEL PROTECTION/
PERSONNEL PROTECTION + FIRE PROTECTION INSULATION
(Resin Bonded Rockwool)**

Nominal size		Insulation Thickness For Temperatures °C							
Inch	mm	100 °C	150 °C	200 °C	250 °C	300 °C	350 °C	400 °C	450 °C
1/2	15	30	30	30	30	30	40	40	50
3/4	20	30	30	30	30	30	40	40	50
1	25	30	30	30	30	30	40	40	50
1 1/4	32	30	30	30	30	30	40	50	50
1 1/2	40	30	30	30	30	30	40	50	60
2	50	30	30	30	30	40	40	50	60
2 1/2	65	30	30	30	30	40	40	50	60
3	80	30	30	30	30	40	50	50	60
3 1/2	95	30	30	30	30	40	50	50	60
4	100	30	30	30	30	40	50	60	70
5	125	30	30	30	30	40	50	60	70
6	150	30	30	30	30	40	50	60	70
8	200	30	30	30	30	40	50	60	70
10	250	30	30	30	30	40	50	60	70
12	300	30	30	30	30	40	50	60	80
14	350	30	30	30	30	40	50	60	80
16	400	30	30	30	30	40	50	70	80
18	450	30	30	30	30	40	50	70	80
20	500	30	30	30	30	40	60	70	80
24	600	30	30	30	30	40	60	70	80
30	750	30	30	30	30	40	60	70	80
36	900	30	30	30	30	40	60	70	80
48	1200	30	30	30	30	40	60	70	80
Above 48	> 1200	30	30	30	30	40	60	70	80

Note: All thickness in mm only.

Basis: Ambient temperature : 25°C

Wind velocity : 8 kmph

Insulation material : Option 1 : Resin bonded rockwool as per IS-8183 for mattresses and IS-9842 for preformed pipe sections

Option 2 : Metal Guards as per UN 2004-06 Part 1,2,3

Noted

(Metal Guard to be specified in area where adequate air circulation is there. Same shall be decided by Contractor based on location of the Equipment)

Project Green-PVC	Client Mundra Petrochem Ltd.	Plant Code MPL-OEPMC	Document Identification Number OEPMC-EQS-0000-EC-00008	Contract No. 66-6753
	TECHNICAL SPECIFICATION FOR THERMAL INSULATION			
				Rev 05 Page 15 of 16

	Cladding material	:	Aluminium (For Personal Protection Insulation)
			Galvanized Steel (For Personal Protection + Fire Protection Insulation)
	Mean thermal conductivity values considered from IS-8183 (for material density 120 - 150 kg/m ³).		
	Conductivity (mw/cm°C at external fire temperature of 904°C) : not higher than 5.77.		
	Flammability (Fire Protection case) : Non-combustible (ASTM E 136 or DIN 4102 or BS 476)		
	The insulating materials and adhesives, which will be used, shall be certified as non-combustible (Class 0-25 to ASTM E84 or equivalent).		
	No. of layers of insulation	:	First layer upto 75 mm
			Second layer > 75 mm to 150 mm
			Third layer > 150 mm

Surface temperature restricted to 55°C maximum calculated using following eqn.

$$(T_p - T_s) / [r_2 \times (\ln r_2/r_1)/K_m] = 0.548 \times \varepsilon \times [(T_s/55.55)^4 - (T_a/55.55)^4] + 1.957 (t_s - t_a)^{1.25} \times (1 + 2.857 V)^{0.5}$$

Tp	: Process Temperature, °K	Ts	: Surface Temperature, °K
Ta	: Ambient Temperature, °K	ts	: Surface Temperature, °C
ta	: Ambient Temperature, °C	r1	: Outer radius of pipe, in mts
r2	: Outer radius of pipe including insulation, in metres		
km	: Mean thermal conductivity W/m°K		
ε	: Surface emittance		
V	: Wind velocity, m/s		

Noted

Project Green-PVC	Client Mundra Petrochem Ltd.	Plant Code MPL-OEPMC	Document Identification Number OEPMC-EQS-0000-EC-00008	Contract No. 66-6753
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Annexure-1 : Reference Standard – STATIC EQUIPMENTS

(Refer Document : OEPMC-EQS-0000-EC-00055)

Standard No.	Part	Title	Issue Dt
UN 9256-01	1 (M)	ERECTION : THERMAL INSULATION	08-2008
UN 5001-02	1 (M)	INSULATION : HEAT AND COLD INSULATION; ARRANGEMENT DETAILS	08-2008
UN 9250-01	1 (M)	ERECTION : WAREHOUSING	08-2008
UN 2004-06 (M)	-	VESSELS & EQUIPMENT: INSULATION CLIPS FOR VERTICAL STEEL VESSELS	02-2016
UN 2004-06	1	VESSELS & EQUIPMENT: PERSONNEL PROTECTION METAL GUARD SUPPORTS FOR VERTICAL VESSEL	07-2011
UN 2004-06	2	VESSELS & EQUIPMENT: PERSONNEL PROTECTION METAL GUARD SUPPORTS FOR VERTICAL VESSEL (AT AREA OF LADDER)	07-2011
UN 2004-06	3	VESSELS & EQUIPMENT: PERSONNEL PROTECTION METAL GUARD SUPPORTS FOR HORIZONTAL VESSEL	07-2011

Noted

		Plant Green-PVC	Customer Mundra Petrochem Ltd.	Code MPL-OEPMC	Contract No.	66-6753	Page 1/3														
APPENDIX – A1 TECHNICAL SPECIFICATION FOR INSULATION																					
1.	TYPE OF INSULATION : ☒ Hot ☐ Cold ☐ Cold + Hot																				
2.	☒ With Electric Tracing ☒ Without Electric Tracing ☒ Fire Protection																				
3.	☒ With Steam Tracing ☒ Without Steam Tracing ☒ Personnel Protection																				
4.	PRE-TREATMENT OF SURFACES TO BE INSULATED :																				
5.	☐ Pre-treatment not required ☒ Pre-treatment according to technical specification for PAINTING																				
6.																					
7.	INSULATION MATERIAL : Resin bonded Rockwool																				
8.	Type of insulation material :				WIRE MESH :																
9.	☒ Mattresses ☒ Slabs				☒ One side ☐ Both sides																
10.	☒ Preformed sections ☐ Insitu foaming				Material :																
11.	Insulation material according to :				☒ G.I. ☐ S.S.																
12.	☒ Mattresses & Slabs - IS 8183 : 1993				Type :																
13.	☒ Preformed pipe sections - IS 9842 : 1994				☒ Hexagonal ☐ Square																
14.	☒ Loose Rockwool - IS 3677 :1985 Type 1				Mesh Size : ¾ inch Dia. Of wire : 24 SWG																
15.					Stitching wire for mattresses and slabs :																
16.	Density : Tolerance				Material : G.I. Dia. : 0.4 mm																
17.	1) Mattresses & Slabs : 120 Kg/m³ ± 15 % when tested in accordance with IS 3144 : 1992																				
18.	2) Preformed pipe sections : 150 Kg/m³ ± 15 % when tested in accordance with IS 5688 : 1982																				
19.					NO. OF LAYERS OF INSULATION :																
20.	Chloride ion content : 12 mg/kg (Max.)				Upto 75 mm - First Layer																
21.	Service Temp. : 550 °C				> 75 mm to 150 mm – Second Layer																
22.					> 150 mm - Third Layer																
23.																					
24.	Thermal conductivity (K) at mean temp. :																				
25.																					
26.	<table border="1"> <tr> <td>Mean Temp. °C</td> <td>50</td> <td>100</td> <td>150</td> <td>200</td> <td>250</td> <td>300</td> </tr> <tr> <td>K - mw/cm °C</td> <td>0.43</td> <td>0.52</td> <td>0.62</td> <td>0.73</td> <td>0.84</td> <td>0.95</td> </tr> </table>							Mean Temp. °C	50	100	150	200	250	300	K - mw/cm °C	0.43	0.52	0.62	0.73	0.84	0.95
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K - mw/cm °C	0.43	0.52	0.62	0.73	0.84	0.95															
27.	Determination of thermal conductivity : As per IS 3346 : 1980																				
28.																					
29.																					
30.																					
31.	Applicable thickness of insulation : Refer insulation table																				
32.																					
33.	NOTES :																				
34.	1) Preformed pipe sections shall be used on pipe sizes upto 8 inch NB and insulation thickness upto 75 mm.																				
35.	2) For pipe sections above 8 inch NB or insulation thickness above 75 mm, mattresses should be used.																				
36.	3) Loose Rockwool shall only be used for filling gaps and insulation on intricate shapes.																				
37.	4) All S.S. and non-ferrous items shall be wrapped with 0.1 mm thick Aluminium foil before application of insulation.																				
38.	5) Following are the tolerances on material properties when tested in accordance with IS 3144 : 1992																				
39.	a) Max. permissible binder content : 3 %																				
40.	b) Max. permissible moisture content : 2 %																				
41.	c) Max. permissible moisture absorption : 2 %																				
42.	d) Max. permissible Sulphur content : 0.6 %																				
43.	6) Slagwool is not acceptable.																				
44.	7) The insulation material shall be rated as incombustible & tested in accordance with IS 3144 : 1992.																				
45.	8) All standards referred in this specification are latest editions.																				
46.	9) Jacketing material shall be Aluminium as specified on page 2/3 except Fire Protection Insulation.																				
47.	10) Fire Protection Jacketing material shall be Galvanized Steel as per page 3/3.																				
48.																					
49.																					
50.																					
51.																					
52.																					
53.																					
54.																					
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		Plant Green-PVC	Customer Mundra Petrochem Ltd.	Code MPL-OEPMC	Contract No.	66-6753	Page 2/3																																									
APPENDIX – A1 TECHNICAL SPECIFICATION FOR INSULATION																																																
1.	CLADDING MATERIAL : Aluminium																																															
2.	Cladding material according to :																																															
3.	1) Plain sheet : IS 737 : 2008 condition H2			2) Circular corrugated : IS 1254 : 1991 condition H3 / H4																																												
4.	Designation : 40800 / 19000 / 31000			Designation : 40800 / 31000																																												
5.	THICKNESS OF CLADDING SHEETS DEPENDING UPON DIAMETER OVER INSULATION :																																															
6.	<table><tr><th>Type of item</th><th>Diameter over Insulation (mm)</th><th>Thk.</th><th>Type of cladding material</th></tr><tr><td>All pipes, fittings, flanges, valves</td><td>Upto 500</td><td>24 SWG</td><td rowspan="3">Plain</td></tr><tr><td>Horizontal equipments & Vertical</td><td>> 500 to 2000</td><td>22 SWG</td></tr><tr><td>Equipment heads</td><td>> 2000</td><td>20 SWG</td></tr><tr><td>Shell of vertical Equipments</td><td>Upto 500</td><td>24 SWG</td><td>Plain</td><td rowspan="3">Circular Corrugation : Pitch – 75 mm Depth – 19 mm</td></tr><tr><td></td><td>> 500 to 1000</td><td>22 SWG</td><td>Plain</td></tr><tr><td></td><td>> 1000</td><td>22 SWG</td><td>Circular Corrugated</td></tr><tr><td>Tank roof</td><td>--</td><td>18 SWG</td><td>Plain</td></tr><tr><td>Horton sphere</td><td>Upto 5000</td><td>22 SWG</td><td rowspan="3">Plain</td></tr><tr><td></td><td>> 5000 to 10000</td><td>20 SWG</td></tr><tr><td></td><td>> 10000</td><td>18 SWG</td></tr></table>							Type of item	Diameter over Insulation (mm)	Thk.	Type of cladding material	All pipes, fittings, flanges, valves	Upto 500	24 SWG	Plain	Horizontal equipments & Vertical	> 500 to 2000	22 SWG	Equipment heads	> 2000	20 SWG	Shell of vertical Equipments	Upto 500	24 SWG	Plain	Circular Corrugation : Pitch – 75 mm Depth – 19 mm		> 500 to 1000	22 SWG	Plain		> 1000	22 SWG	Circular Corrugated	Tank roof	--	18 SWG	Plain	Horton sphere	Upto 5000	22 SWG	Plain		> 5000 to 10000	20 SWG		> 10000	18 SWG
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49.	METAL SEALANT				THERMAL SHIELD :																																											
50.	Requirement	: Elastic, Fire resistant	Material : Alumunium Foil																																													
51.	Recommendation	: Loid Seal 94 or MAS 94 or equi.	Thk. : 0.1 mm																																													
52.	Note : Floating rings for insulation supports on vessel heads shall be ¼ " dia., S.S. when exposed else C.S. rods with weded ends.																																															
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APPENDIX – A1 TECHNICAL SPECIFICATION FOR INSULATION																																													
1.	CLADDING MATERIAL : Galvanised steel																																												
2.	Galvanised steel sheets shall be made by galvanising cold rolled sheets or coils.																																												
3.	Cladding material according to :																																												
4.	1) Plain sheet / Circular corrugated : IS 277 : 2018, with 275gm/m2 of coating																																												
5.	THICKNESS OF CLADDING SHEETS DEPENDING UPON DIAMETER OVER INSULATION :																																												
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APPENDIX – A3 TECHNICAL SPECIFICATION FOR INSULATION							
1.	TYPE OF INSULATION : ☐ Hot ☒ Cold ☐ Cold + Hot						
2.	☐ With Electric Tracing ☐ Without Electric Tracing ☒ Anti-Sweat						
3.	☐ With Steam Tracing ☐ Without Steam Tracing ☐ Personnel Protection						
4.	PRE-TREATMENT OF SURFACES TO BE INSULATED :						
5.	☐ Pre-treatment not required ☒ Pre-treatment according to technical specification for PAINTING						
6.							
7.	INSULATION MATERIAL : Polyurethane Foam						
8.	Type of insulation material :				NO. OF LAYERS OF INSULATION		
9.	☐ Mattresses ☒ Slabs				Upto 50 mm - First Layer		
10.	☒ Preformed sections ☐ Insitu foaming				> 50 mm to 100 mm - Second Layer		
11.	Insulation material according to :				> 100 mm - Third Layer		
12.	☒ Slabs - IS 12436 Type 1				ADHESIVE :		
13.	☒ Preformed pipe sections - IS 12436 Type 1				Purpose : 1) For fixing insulation on items		
14.					2) For joining adjacent slabs / sections		
15.					Material :		
16.					☒ Loid Bond 17 / MAS 83 or equivalent		
17.	Density : Tolerance				☐		
18.	1) Slabs : 34 Kg/m ³ ± 5 %				Thickness : 1.0 mm approx.		
19.	2) Preformed pipe sections : 34 Kg/m ³ ± 5 %				Manufacturer's instructions regarding application of Adhesive		
20.					Shall be followed.		
21.	Chloride ion content : 0 mg/kg (Max.)						
22.	Temp. resistance : -150 °C upto +110 °C						
23.							
24.	Thermal conductivity (K) at mean temp. :						
25.							
26.							
27.							
28.							
29.							
30.							
31.	Compressive strength : 2 - 3 kg/cm ²						
32.	Applicable thickness of insulation : Refer insulation table						
33.							
34.	VAPOUR BARRIER :						
35.	LB 130 / MAS 130 or equivalent, fire resistive aluminium mastic alongwith fibreglass woven cloth.						
36.	Thickness : 4 mm. approx.						
37.							
38.							
39.	NOTES :						
40.	1) The gaps between adjacent slabs / sections shall be kept as minimum as possible. Larger unavoidable gaps must be filled						
41.	with foam sections that are cut to suit.						
42.	2) Mineral wool shall be used to fill expansion joints & other hollow spacers on valves etc. This mineral wool shall conform to						
43.	IS 3677 Type 1 / IS 3690.						
44.	3) In case of equipment, slabs shall be cut into 125 to 250 mm widths (depending on dia.) and then fitted to the curvature.						
45.	4) Wherever applicable, care must be taken that the transition between steel & vapour barrier / cladding is properly sealed with						
46.	Loid Seal 94 / LB 130 / MAS 130 or equivalent.						
47.	5) Insulation work whereby sheet metal jackets are used must be carried out without employing self tapping screws as these						
48.	would penetrate vapour barrier & render the same useless. (Not applicable for insitu foaming)						
49.	6) The insulation material shall be rated as incombustible & tested in accordance with relevant IS code.						
50.	7) All standards referred in this specification are latest editions.						
51.							
52.							
53.							
54.							
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1.	CLADDING MATERIAL : Aluminium																																												
2.	Cladding material according to :																																												
3.	1) Plain sheet : IS 737: 2008 condition H2			2) Circular corrugated : IS 277 condition H3 / H																																									
4.	Designation : 40800 / 19000 / 31000			Designation : 40800 / 31000																																									
5.	THICKNESS OF CLADDING SHEETS DEPENDING UPON DIAMETER OVER INSULATION :																																												
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1.	CLADDING MATERIAL : Galvanised steel																																												
2.	Galvanised steel sheets shall be made by galvanising cold rolled sheets or coils.																																												
3.	Cladding material according to :																																												
4.	1) Plain sheet / Circular corrugated : IS 277 : 2018 with 275gm/m2 of coating																																												
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1.	TYPE OF INSULATION : ☐ Hot ☒ Cold ☐ Cold + Hot ☒ Fire Protection						
2.	☐ With Electric Tracing ☐ Without Electric Tracing ☒ Anti-Sweat						
3.	☐ With Steam Tracing ☐ Without Steam Tracing ☐ Personnel Protection						
4.	PRE-TREATMENT OF SURFACES TO BE INSULATED :						
5.	☐ Pre-treatment not required ☒ Pre-treatment according to technical specification for PAINTING						
6.							
7.	INSULATION MATERIAL : Cellular Glass (Foam Glass)						
8.	Type of insulation material :				NO. OF LAYERS OF INSULATION		
9.	☐ Mattresses ☒ Slabs				Upto 50 mm - First Layer		
10.	☒ Preformed sections ☐ Insitu foaming				> 50 mm to 100 mm - Second Layer		
11.	Insulation material according to :				> 100 mm - Third Layer		
12.	☒ Slabs - ASTM C552				ADHESIVE :		
13.	☒ Preformed pipe sections - ASTM C552				Purpose : 1) For fixing insulation on items		
14.					2) For joining adjacent slabs / sections		
15.					Material :		
16.					☒ Loid Bond 17 / MAS 83 or equivalent		
17.	Density:98-139 Kg/m ³ Tolerance				☐		
18.	1) Slabs : Acc. To ASTM C303/302				Thickness : 1.0 mm approx.		
19.	2) Preformed pipe sections : Acc. To ASTM C303/302				Manufacturer's instructions regarding application of Adhesive Shall be followed.		
20.							
21.	Chloride ion content : 0 mg/kg (Max.)						
22.	Service Temp. : -268°C to 427°C						
23.							
24.	Thermal conductivity (K) at mean temp. :						
25.							
26.							
27.							
28.							
29.							
30.							
31.	Compressive strength : As per ASTM Code						
32.	Applicable thickness of insulation : Refer insulation table						
33.							
34.	VAPOUR BARRIER :						
35.	LB 130 / MAS 130 or equivalent, fire resistive aluminium mastic along with fiberglass woven cloth.						
36.	Thickness : 4 mm. approx.						
37.	Aluminium foil reinforced with Vinyl and Scrim fire resistive vapour barrier film with joints sealed with self adhesive tapes						
38.	of same base.						
39.							
40.	NOTES :						
41.	1) The gaps between adjacent slabs / sections shall be kept as minimum as possible. Larger unavoidable gaps must be filled						
42.	with foam sections that are cut to suit.						
43.	2) Glass wool shall be used to fill expansion joints & other hollow spacers on valves etc. This glass wool shall conform to						
44.	ASTM C592.						
45.	3) In case of equipment, slabs shall be cut into 125 to 250 mm widths (depending on dia.) and then fitted to the curvature.						
46.	4) Wherever applicable, care must be taken that the transition between steel & vapour barrier / cladding is properly sealed with						
47.	Loid bond 17 mixed with cellular glass powder.						
48.	5) Insulation work whereby sheet metal jackets are used must be carried out without employing self tapping screws as these						
49.	would penetrate vapour barrier & render the same useless. (Not applicable for insitu foaming)						
50.	Hence self tapping screws shall not be used for cold insulation.						
51.	6) The insulation material shall be rated as incombustible & tested in accordance with ASTM E136, ASTM E84.						
52.	7) All standards referred in this specification are latest editions.						
53.	8) Foamglass to be applied in min 2 layers at pipelines, fittings and equipment; the thickness of the layers shall be similar;						
54.	at valves and flanges 1 layer is acceptable.						
55.	9) Manufacturer to follow other applicable requirements as per UL 1709 std.						
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Noted

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2.	Galvanised steel sheets shall be made by galvanising cold rolled sheets or coils.																																												
3.	Cladding material according to :																																												
4.	1) Plain sheet / Circular corrugated : IS 277 : 2018 with 275gm/m2 of coating																																												
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